Team Problems

1. Let $1.\overline{81} = 1.81818181\ldots = \dfrac{N}{D}$, where N and D are two positive integers. Compute the minimum value of $N + D$.

2. A two-digit number N is chosen at random, with all two-digit numbers equally likely. Compute the probability that at least one of $\log_2 N$, $\log_3 N$, $\log_5 N$, or $\log_7 N$ is an integer.

3. The difference between the areas of the circumcircle and incircle of an equilateral triangle is 27π. Compute the area of the triangle.

4. If Sam arranges his candies in piles of 6, there are 2 left over. If he arranges them in piles of 9, there are 5 left over. If he arranges them in piles of 15, there are 11 left over. Sam has between 100 and 200 candies. Compute how many candies Sam has.

5. Quadrilateral $JANE$ is inscribed in a circle. Diameter JN has length 50, $AN = 48$ and $NE = 30$. Compute the perimeter of $JANE$.

6. A fair coin is flipped repeatedly and the throws are recorded until three consecutive flips are (in order) Heads, Heads, and Tails *or* Heads, Tails, and Tails. Compute the probability that the sequence Heads, Heads, and Tails appears first.

7. Let a, b, c, and d be real numbers such that the graphs of $y = -|x - a| + b$ and $y = |x - c| + d$ intersect at $(2, 1)$ and $(8, -1)$. Compute $a + c$.

8. Given triangle ABC with $AB = 6, BC = 8$, and $AC = 7$. Point D is on AC such that BD bisects angle B. Circles C_1 and C_2 are drawn such that C_1 is internally tangent to each side of triangle ABD, and C_2 is internally tangent to each side of triangle CBD. Compute the sum of the areas of circles C_1 and C_2.

9. Compute $\dfrac{1}{1 \times 3} + \dfrac{1}{3 \times 5} + \cdots + \dfrac{1}{2007 \times 2009} + \dfrac{1}{2009 \times 2011} = \displaystyle\sum_{n=1}^{1005} \frac{1}{(2n-1)(2n+1)}$.

10. An *ordered partition* of an integer N is a sequence of positive integers that sum to N. As an example, $1 + 3$, $2 + 1 + 1$, and $1 + 1 + 2$ are distinct ordered partitions of 4. Compute the number of ordered partitions of 10 that consist of only odd positive integers.